

Influence of halide composition on the structural, electronic, and optical properties of mixed $\text{CH}_3\text{NH}_3\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$ perovskites calculated using the virtual crystal ...

UG Jong, CJ **Yu**, JS Ri, NH Kim, GC Ri - Physical review B, 2016 - APS

Extensive studies have demonstrated the promising capability of the organic-inorganic hybrid halide perovskite $\text{CH}_3\text{NH}_3\text{PbI}_3$ in solar cells with a high power conversion efficiency ...

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Advances in modelling and simulation of halide perovskites for solar cell applications

CJ **Yu** - Journal of Physics: Energy, 2019 - iopscience.iop.org

Perovskite solar cells (PSCs) are attracting much attention as the most promising candidate for the next generation of solar cells. This is due to their low cost and high power conversion ...

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First-principles study on structural, electronic and optical properties of halide double perovskite Cs_2AgBX_6 (B= In, Sb; X= F, Cl, Br, I)

CJ **Yu**, IC Ri, HM Ri, JH Jang, YS Kim, UG Jong - RSC advances, 2023 - pubs.rsc.org

All-inorganic halide double perovskites (HDPs) attract significant attention in the field of perovskite solar cells (PSCs) and light-emitting diodes. In this work, we present a first-principles ...

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Influence of water intercalation and hydration on chemical decomposition and ion transport in methylammonium lead halide perovskites

UG Jong, CJ **Yu**, GC Ri, AP McMahon... - Journal of Materials ..., 2018 - pubs.rsc.org

The application of methylammonium (MA) lead halide perovskites, $\text{CH}_3\text{NH}_3\text{PbX}_3$ (X = I, Br, Cl), in perovskite solar cells has made great recent progress in performance efficiency ...

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Critical role of water in defect aggregation and chemical degradation of perovskite solar cells

YH Kye, CJ **Yu**, UG Jong, Y Chen... - The journal of physical ..., 2018 - ACS Publications

The chemical stability of methylammonium lead iodide (MAPbI_3) under humid conditions remains the primary challenge facing halide perovskite solar cells. We investigate defect ...

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First-principles study on structural, electronic, and optical properties of inorganic Ge-based halide perovskites

UG Jong, CJ **Yu**, YH Kye, YG Choe, W Hao... - Inorganic ..., 2019 - ACS Publications

Using density functional theory calculations, we explore the structural, electronic, and optical properties of the inorganic Ge-based halide perovskites AGeX_3 (A = Cs, Rb; X = I, Br, Cl) ...

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Computational prediction of structural, electronic, and optical properties and phase stability of double perovskites K_2SnX_6 (X= I, Br, Cl)

UG Jong, CJ **Yu**, YH Kye - RSC advances, 2020 - pubs.rsc.org

The vacancy-ordered double perovskites K_2SnX_6 (X = I, Br, Cl) attract significant research interest due to their potential applications as light absorbing materials in perovskite solar cells...

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First-principles study on the material properties of the inorganic perovskite $\text{Rb}_{1-x}\text{Cs}_x\text{PbI}_3$ for solar cell applications

UG Jong, CJ **Yu**, YS Kim, YH Kye, CH Kim - Physical Review B, 2018 - APS

Recently, replacing or mixing organic molecules in hybrid halide perovskites with inorganic Cs or Rb cations was reported to increase the material stability with comparable solar cell ...

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Revealing the stability and efficiency enhancement in mixed halide perovskites $\text{MAPb}(\text{I}_{1-x}\text{Cl}_x)_3$ with ab initio calculations

UG Jong, CJ **Yu**, YM Jang, GC Ri, SN Hong... - Journal of Power ..., 2017 - Elsevier

A little addition of Cl to MAPbI_3 has been reported to improve the material stability as well as light harvesting and carrier conducting properties of organometal trihalide perovskites, the ...

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First-principles study on the optoelectronic and mechanical properties of all-inorganic lead-free fluoride perovskites ABF_3 (A= Na, K and B= Si, Ge)

..., UG Jong, CJ Kang, YS Kim, YH Kye, CJ **Yu** - Materials ..., 2023 - pubs.rsc.org

In spite of extensive studies on halide perovskites for advanced photovoltaic applications, little attention has been paid to fluoride perovskites so far. Here, we present a systematic ...

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Vacancy-Driven Stabilization of the Cubic Perovskite Polymorph of CsPbI₃

YH Kye, CJ **Yu**, UG Jong, KC Ri, JS Kim... - The Journal of ..., 2019 - ACS Publications

The inorganic halide perovskite CsPbI₃ has shown great promise for efficient solar cells, but the instability of its cubic phase remains a major challenge. We present a route for ...

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Preparation of acrylic emulsion coating with melamine polyphosphate, pentaerythritol and titanium dioxide for flame retardant cotton/polyethylene terephthalate blend ...

CH Jo, YM Jang, DH Mun, CJ **Yu**, CG Choe... - Polymer Degradation And ..., 2023 - Elsevier

An organic phosphorus-nitrogen containing flame retardant melamine polyphosphate (MPP) was synthesized with two-step approach using melamine and phosphoric acid, and ...

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Anharmonic phonons and phase transitions in the vacancy-ordered double perovskite Cs₂SnI₆ from first-principles predictions

UG Jong, CJ **Yu**, YH Kye, SH Choe, JS Kim, YG Choe - Physical Review B, 2019 - APS

Tin-based vacancy-ordered double perovskite Cs₂SnI₆ has emerged as a potential candidate for nontoxic, nonhygroscopic, and high efficient light absorber in the field of photovoltaics...

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Manifestation of the thermoelectric properties in Ge-based halide perovskites

UG Jong, CJ **Yu**, YH Kye, SN Hong, HG Kim - Physical Review Materials, 2020 - APS

In spite of intensive studies on the chalcogenides as conventional thermoelectrics, it remains a challenge to find a proper material with high electrical but low thermal conductivities. In ...

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A first-principles study on the chemical stability of inorganic perovskite solid solutions Cs_{1-x}Rb_xPbI₃ at finite temperature and pressure

UG Jong, CJ **Yu**, YH Kye, YS Kim, CH Kim... - Journal of Materials ..., 2018 - pubs.rsc.org

The inorganic halide perovskite Cs(Rb)PbI₃ has attracted significant research interest for its application as a light-absorbing material in perovskite solar cells (PSCs). Although there ...

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High Thermoelectric Performance in the Cubic Inorganic Cesium Iodide Perovskites CsBI₃ (B = Pb, Sn, and Ge) from First-Principles

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Searching thermoelectric materials with high performance and low cost is now receiving special attention and great challenges in the field of material design. In this work, we perform first...

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Ab initio Investigation of Adsorption Characteristics of Bisphosphonates on Hydroxyapatite (001) Surface

MH Ri, YM Jang, US Ri, CJ **Yu**, KI Kim... - Journal of Materials ..., 2018 - Springer

Structures of some bisphosphonates (clodronate, etidronate, pamidronate, alendronate, risedronate, zoledronate) were relaxed and analyzed by DFT method. By comparing their ...

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First-principles study of ternary graphite compounds cointercalated with alkali atoms (Li, Na, and K) and alkylamines towards alkali ion battery applications

GC Ri, CJ **Yu**, JS Kim, SN Hong, UG Jong... - Journal of Power Sources, 2016 - Elsevier

First-principles calculations were carried out to investigate the structural, energetic, and electronic properties of ternary graphite compounds cointercalated with alkali atoms (AM = Li, Na...

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Structural and electrochemical trends in mixed manganese oxides Na_x(M_{0.44}Mn_{0.56})O₂ (M= Mn, Fe, Co, Ni) for sodium-ion battery cathode

CJ **Yu**, YC Pak, CH Kim, JS Kim, KC Ri, KH Ri... - Journal of Power ..., 2021 - Elsevier

Developing cost-effective and high performance sodium-ion batteries (SIBs) relies mostly on advanced cathode materials with high electrode voltage, high capacity and fast sodium-ion ...

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Point defects and their impact on electrochemical performance in Na_{0.44}MnO₂ for sodium-ion battery cathode application

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Sodium manganese oxide Na_{0.44}MnO₂ (NMO) in an open structure with large tunnels is of great interest for sodium-ion battery cathode materials due to its high electrode voltage and ...

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Interface Engineering in Hybrid Iodide CH₃NH₃PbI₃ Perovskites Using Lewis Base and Graphene toward High-Performance Solar Cells

CJ **Yu**, YH Kye, UG Jong, KC Ri, SH Choe... - ... Applied Materials & ..., 2019 - ACS Publications

Photovoltaic solar cells based on organic–inorganic hybrid halide perovskites have achieved a substantial breakthrough via advanced interface engineering. Reports have emphasized ...

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Ab initio study of sodium cointercalation with diglyme molecule into graphite

CJ **Yu**, SB Ri, SH Choe, GC Ri, YH Kye, SC Kim - Electrochimica Acta, 2017 - Elsevier

The cointercalation of sodium with the solvent organic molecule diglyme into graphite, forming a ternary graphite intercalation compound (t-GIC), can resolve the difficulty of making the ...

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Antioxidation activity of molecular hydrogen via protoheme catalysis in vivo: an insight from ab initio calculations

SA Kim, YC Jong, MS Kang, CJ **Yu** - Journal of Molecular Modeling, 2022 - Springer

Recently, molecular hydrogen has been found to exhibit antioxidation activity through many clinical experiments, but the mechanism has not been fully understandable at atomic level. ...

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Effect of porosity and temperature on thermal conductivity of jennite: A molecular dynamics study

SN Hong, CJ **Yu**, US Hwang, CH Kim, BH Ri - Materials Chemistry and ..., 2020 - Elsevier

Despite growing interest of cement materials in design and construction of green buildings, their thermal properties are not well understood at atomic scale. In this work, we reported the ...

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Preparation and characterization of durable flame retardant cotton fabrics using phosphorylation reaction of cellulose with ammonium etidronate

..., WP Kung, YM Jang, P Choe, CG Choe, CJ **Yu** - Iranian Polymer ..., 2024 - Springer

Developing an effective flame retardant for cotton is urgently needed to minimize the life and property loss caused by fire hazard. Here, diphosphonate system flame retardant, the ...

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Interfacial Enhancement of Photovoltaic Performance in MAPbI₃/CsPbI₃ Superlattice

..., UH Ko, YH Kye, UG Jong, CJ **Yu** - ACS Applied Materials ..., 2021 - ACS Publications

Perovskite solar cells have continued to fascinate over the past decade due to fast increasing power conversion efficiency and very low fabrication cost but still suffered from poor ...

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Ab initio thermodynamic study of the SnO₂ (110) surface in an O₂ and NO environment: A fundamental understanding of the gas sensing mechanism for NO and NO ...

SN Hong, YH Kye, CJ **Yu**, UG Jong, GC Ri... - Physical Chemistry ..., 2016 - pubs.rsc.org

For the purpose of elucidating the gas sensing mechanism of SnO₂ for NO and NO₂ gases, we determine the phase diagram of the SnO₂(110) surface in contact with an O₂ and NO ...

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Revealing the influence of porosity and temperature on transport properties of nanobubble solution with molecular dynamics simulations

SN Hong, JH Ri, SY Mun, CJ **Yu** - Journal of Molecular Liquids, 2022 - Elsevier

Viscosity reduction and diffusivity increase of aqueous solution is of great importance in many fluid-related industrial processes. To achieve the purpose in environment-friendly and cost-...

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Defect formation and ambivalent effects on electrochemical performance in layered sodium titanate Na₂Ti₃O₇

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Point defects can be formed readily in layered transition metal oxides used as electrode materials for alkali-ion batteries but their influence on the electrode performance is yet obscure. ...

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All-inorganic halide perovskites have attracted a great interest as a promising light harvester of perovskite solar cells due to their enhanced chemical stability. In this work we investigate ...

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Compressive strain-induced enhancement of thermoelectric performance in lead-free halide double perovskites K2SnX6 (X= I, Br, Cl)

..., SH Kim, RW Ham, S Ri, RJ Ri, CJ Yu - Applied Physics ..., 2024 - pubs.aip.org

Exploring thermoelectric materials with high performance and low cost is of great importance in mitigating environmental and energy challenges. Here, we provide an atomistic insight ...

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Twofold rattling mode-induced ultralow thermal conductivity in vacancy-ordered double perovskite Cs 2 SnI 6

..., CH Ri, YH Kye, CJ Pak, S Cottenier, CJ Yu - Chemical ..., 2022 - pubs.rsc.org

We report a first-principles study of lattice vibrations and thermal transport in Cs2SnI6, the vacancy-ordered double perovskite. Twofold rattlers of Cs atoms and SnI6 clusters in Cs2SnI6, ...

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An efficient virtual crystal approximation that can be used to treat heterovalent atoms, applied to (1– x) BiScO3– xPbTiO3

CJ Yu, H Emmerich - Journal of Physics: Condensed Matter, 2007 - iopscience.iop.org

It is well accepted that the virtual crystal approximation provides an effective method for studying solid solutions and alloys from first principles. Here we propose another approach ...

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Preparation and flame retardant properties of cotton fabrics treated with resorcinol bis (diphenyl phosphate)

YM Jang, CJ Yu, KS Choe, CH Choe, CH Kim - Cellulose, 2021 - Springer

A flame-retardant agent, resorcinol bis(diphenyl phosphate) (RDP), was synthesized with resorcinol, phenol and phosphorus oxychloride under a catalytic action of magnesium chloride...

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CH Ri, YS Kim, UG Jong, YH Kye, SH Ryang, CJ Yu - RSC advances, 2021 - pubs.rsc.org

Perovskite materials have been recently attracting a great amount of attention as new potential photocatalysts for water splitting hydrogen evolution. Here, we propose lead-free ...

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... **Chol-Jun Yu** developed the original project. Yun-Kyong Ri, Song-Ae Kim and **Chol-Jun Yu** performed the calculations and drafted the first manuscript. Yun-Hyok Kye, **Yu**-Chol Jong and ...

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Role of water molecules in enhancing the proton conductivity on reduced graphene oxide under high humidity

GC Ri, JS Kim, CJ Yu - Physical Review Applied, 2018 - APS

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Molecular dynamics study of the effect of moisture and porosity on thermal conductivity of tobermorite 14 Å

SN Hong, CJ Yu, KC Ri, JM Han, BH Ri - International Journal of Thermal ..., 2021 - Elsevier

Understanding of thermal properties of building materials is of great importance in design of buildings with efficient energy saving. In this work, we have studied the effect of moisture ...

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Developing highly efficient photocatalysts for the hydrogen evolution reaction (HER) by solar-driven water splitting is a great challenge. Here, we study the atomistic origin of interface ...

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Atomic structures and energetics of heterophase interfaces among Fe (100), TiC (110) and Mo 2 C (001) surfaces: insights from first-principles calculations

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... **Chol-Jun Yu** and Chol-Song Pang developed the original project. **Chol-Jun Yu** and Kyong-A Rim performed the calculations and drafted the first manuscript. Chol Ryu, Hyok-Bom Yun, ...
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Graphite has been reported to have anion, as well as cation, intercalation capacities as both a cathode and an anode host material for dual ion batteries. In this work, we study the ...
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Discovering new electrodes for the sodium-ion battery requires a clear understanding of the material process during battery operation. Using first-principles calculations, we identify ...
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Effective cracking of heavy crude oil by using shock-induced nanobubble collapse: A molecular dynamics study

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Ab initio design of drug carriers for zoledronate guest molecule using phosphonated and sulfonated calix [4] arene and calix [4] resorcinarene host molecules

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In the present paper, we provide the refined pressure–temperature phase coexistence line between graphite and diamond with van der Waals (vdW) energy corrected density-functional ...
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Among the recent achievements of sodium-ion battery (SIB) electrode materials, hybridization of two-dimensional (2D) materials is one of the most interesting appointments. In this work, ...
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We investigate the ferrimagnetism and ferroelectricity of bulk NiFe₂O₄ with tetragonal P 4 1 22 symmetry by means of density functional theory calculations using generalized gradient ...
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Effect of porosity and temperature on viscosity and diffusivity of benzene liquid containing nanobubble with molecular dynamics

JH Ri, SN Hong, CH Ri, CJ **Yu** - Fluid Phase Equilibria, 2024 - Elsevier

Introduction of nanobubble (NB) into liquid can significantly affect the transport properties such as viscosity and diffusivity, being important in the design and optimization of many liquid-...

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All-inorganic metal-halide perovskites are demonstrating good prospects toward long-term stability and high efficiency in the field of perovskite photovoltaics. Stemming from CsPbI 3 , ...

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First-principles study of Na x TiO 2 with trigonal bipyramid structures: an insight into sodium-ion battery anode applications

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Developing efficient anode materials with low electrode voltage, high specific capacity and superior rate capability is urgently required on the road to commercially viable sodium-ion ...

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The mechanism of superstability of nanobubbles in liquid confirmed by many experimental studies is still in debate since the classical diffusion predicts their lifetime on the order of a ...

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Superior thermoelectric properties of ternary chalcogenides CsAg 5 Q 3 (Q= Te, Se) predicted using first-principles calculations

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Tailoring novel thermoelectric materials (TEMs) with a high efficiency is challenging due to the difficulty in realizing both low thermal conductivity and high thermopower factor. In this ...

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Metal phosphide CuP 2 as a promising thermoelectric material: an insight from a first-principles study

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In the search for better thermoelectric materials, metal phosphides have not been considered to be viable candidates so far, due to their large lattice thermal conductivity. Here we study ...

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Effect of vacancy concentration on the lattice thermal conductivity of CH 3 NH 3 Pbl 3: a molecular dynamics study

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First-principles study of ethylene carbonate adsorption on prismatic hard carbon surface: An insight into solid-electrolyte interphase formation

C Ryu, JS Kim, SB Rim, SH Choe, CJ **Yu** - Applied Surface Science, 2022 - Elsevier

Solid-electrolyte interphase (SEI) formation at the hard carbon-based anode of sodium-ion batteries (SIBs) is a key process determining the cycling performance of batteries. In this work...

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Harmonizing mechanical and thermal properties in Al/SiC superlattices: *Ab initio* machine-learning-potential study

CJ Yu, JH Jang, KC Ri, SM Kim, CH Pak, RJ Kim - Physical Review B, 2024 - APS

The aluminum matrix composites (AMCs) reinforced with silicon carbide have attracted significant interest in several high-tech industries. In this work, we provide an insightful atomistic ...

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First-principles study on enhancing the photocatalytic hydrogen evolution performance in Cs₃Bi₂I₉/MoS₂ heterostructure with interfacial defect engineering

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Hydrogen has been attracting continuously growing interest as a highly efficient and clean energy source for replacing fossil fuels in the future, and thus developing highly efficient ...

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Atomistic insight into the promising thermoelectric performance of the copper-based ternary phosphide CaCuP

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A first-principles study on the thermoelectric properties of the copper-based ternary phosphide CaCuP is presented. Self-energy relaxation time approximation and unified theory for ...

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Ab initio investigation of the adsorption of zoledronic acid molecule on hydroxyapatite (001) surface: an atomistic insight of bone protection

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We report a computational study of the adsorption of zoledronic acid molecule on hydroxyapatite (001) surface within ab initio density functional theory. The systematic study has been ...

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GC Ri, SH Choe, CJ Yu - Journal of Power Sources, 2018 - Elsevier

Natural abundance of sodium and its similar behavior to lithium triggered recent extensive studies of cost-effective sodium-ion batteries (SIBs) for large-scale energy storage systems. A ...

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Revealing the catalytic oxidation mechanism of CO on α-Fe₂O₃ surfaces: an ab initio thermodynamic study

..., YH Kye, MI Pang, YM Ho, HC Choe, CJ Yu... - Physical Chemistry ..., 2025 - pubs.rsc.org

Significant research efforts have been devoted to improving the efficiency of catalytic carbon monoxide (CO) oxidation over α-Fe₂O₃-based catalysts, but details of the underlying ...

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Electronic structure and photoabsorption property of pseudocubic perovskites CH₃NH₃PbX₃ (X = I, Br) including van der Waals interaction

CJ Yu, UG Jong, MH Ri, GC Ri, YH Pae - Journal of Materials Science, 2016 - Springer

Using density functional theory with the inclusion of van der Waals (vdW) interaction, we have investigated electronic energy bands, density of states, effective masses of charge carriers...

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A first-principles study of interfacial vacancies in the β-CsPbI₃/1T-MoS₂ heterostructure towards photocatalytic applications

..., O Song-li, CS Jang, YS Kim, JS Kim, CJ Yu - Physical Chemistry ..., 2025 - pubs.rsc.org

... Chol-Hyok Ri and **Chol-Jun Yu** developed ... , **Yu**-Song Kim and Jin-Song Kim assisted with the post-processing of calculation results and contributed to useful discussions. **Chol-Jun Yu** ...

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Exploring the role of MgO nanoparticles in the mechanical properties of the MgO/PE composite: a tight-binding and molecular dynamics simulation study

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Metal oxide nanoparticles have been widely used as reinforcing agents in polyethylene (PE)-based composites but the reinforcing mechanism is not yet fully understood. Here we report ...

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Insight into the structural and electrochemical properties of the interface between a Na₆SOI₂ solid electrolyte and a metallic Na anode

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Solid electrolytes (SE) have attracted a great deal of interest as they can not only mitigate the safety issues related to currently used liquid organic electrolytes but also enable the ...

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Revealing the effect of Nb or V doping on anode performance in Na₂Ti₃O₇ for sodium-ion batteries: a first-principles study

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Sodium titanate Na₂Ti₃O₇ (NTO) has superior electrochemical properties as an anode material in sodium-ion batteries (SIBs), and Nb or V doping is suggested to enhance the ...

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CJ **Yu**, SG Hwang, YC Pak, SH Choe... - The Journal of ..., 2020 - ACS Publications

Sodium titanate Na₂Ti₃O₇ (NTO) is regarded as a highly promising anode material with a very low voltage for Na-ion batteries and capacitors but suffers from relatively low specific ...

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Revealing the flame retardancy of cotton fabrics treated with ammonium ethylenediamine tetramethylenephosphonate and trimethylol melamine

YM Jang, CJ Kim, JY Kim, DH Mun, CJ **Yu** - RSC advances, 2025 - pubs.rsc.org

Phosphorus-nitrogen containing polymers have been attracting considerable interest as potential flame retardants (FR) for cotton fabrics. In this work, we report an experimental-...

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Theoretical insight into potential thermoelectric performance of ternary metal phosphide CaAgP

UG Jong, C Ryu, CJ Kang, CJ **Yu** - Applied Physics Letters, 2023 - pubs.aip.org

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Promising photovoltaic performance in ternary chalcogenides ASbS₂ (A= Na, K): Atomistic insights from ab initio calculations

UG Jong, IC Ri, CJ **Yu** - Applied Physics Letters, 2025 - pubs.aip.org

Ternary chalcogenides (TCGs) are drawing significant attention as a promising light absorber for thin-film solar cells but their material properties are not yet well understood. Here, we ...

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Revealing the origin of promising photovoltaic performance in ternary chalcogenide NaBiS₂ from first principles

UG Jong, CH Jang, JS Kim, CJ **Yu** - Applied Physics Letters, 2025 - pubs.aip.org

Ternary chalcogenide NaBiS₂ has emerged as a promising candidate for light absorbers of solar cells, but its optoelectronic properties are not yet fully understood. Here, we report a ...

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First-principles study on structural, electronic, magnetic and thermodynamic properties of lithium ferrite LiFe₅O₈

SY Kim, KS Kim, UG Jong, CJ Kang, SC Ri, CJ **Yu** - RSC advances, 2022 - pubs.rsc.org

Lithium ferrite, LiFe₅O₈ (LFO), has attracted great attention for various applications, and there has been extensive experimental studies on its material properties and applications. ...

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Fast ionic conductivity by thermal treatment with ultralow electronic transport in solid-state electrolyte Na₃YCl₆

TI Ri, SG Hwang, JS Kim, KC Ri, CJ **Yu** - Applied Physics Letters, 2025 - pubs.aip.org

Improving the ionic and electronic conductivities of solid-state electrolytes (SSEs) is urgently needed to develop commercially viable all-solid-state batteries. Here, we provide atomistic ...

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Insights into low electronic and ultrahigh ionic conductivities of sodium tantalum oxychlorides NaTaOxCl_{6-2x} in crystalline and amorphous phases

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Anion mixing and amorphization are recognized as an effective strategy to develop solid-state electrolytes with low electronic but high ionic conductivities. Here, we present atomistic ...

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Defect properties in Yb 3+-doped CaF 2 from first-principles calculations: a route to defect engineering for up-and down-conversion photoluminescence

YH Kye, CJ **Yu**, UG Jong, CN Sin, W Qin - Journal of Materials ..., 2019 - pubs.rsc.org

Calcium fluoride has been widely used for light up-/down-conversion luminescence by accommodating lanthanide ions as sensitizers or activators. Especially, Yb-doped CaF2 exhibits ...

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Tuning exciton binding and photoabsorption in vacancy-ordered halide double perovskites Rb2ZrX6 (X= Cl, Br, I)

IC Ri, M Choe, SG Hwang, CJ **Yu** - Applied Physics Letters, 2025 - pubs.aip.org

Vacancy-ordered halide double perovskites (VHDPs) have emerged as a promising candidate for broadband luminescence, but the underlying mechanism remains obscure. Here, we ...

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Atomistic insight into enhanced ionic conductivity in mixed halide sodium ion conductor Na3YClxBBr6– x

TI Ri, KC Ri, IJ Kim, JR Hwang, CJ **Yu** - Applied Physics Letters, 2025 - pubs.aip.org

Exploring solid-state electrolytes with high ionic conductivity and low electronic conductivity is of great importance in realizing all-solid-state batteries. Here, we provide an atomistic ...

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Preparation of ammonium ethylenediamine tetramethylenephosphonate and characterization of durable flame-retardant cotton

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Flame-retardant cotton is of great importance in clothing, household goods and various industrial products. In this work, we report a synthesis of phosphorus-rich ammonium ...

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First-principles study of organically modified muscovite mica with ammonium (NH₄⁺) or methylammonium (CH₃NH₃⁺) ion

CJ **Yu**, SH Choe, YM Jang, GH Jang... - Journal of Materials ..., 2016 - Springer

Using density functional theory calculations, we have investigated the interlayer cation exchange phenomena in muscovite mica, which is motivated by a necessity to develop flexible ...

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Formation and characterization of ceramic coating from alumino silicate mineral powders in the matrix of cement composite on the concrete wall

CJ **Yu**, BH Ri, CH Kim, US Hwang, KC Ri... - Materials Chemistry and ..., 2019 - Elsevier

Enhancement of thermal performance of concrete wall is nowadays of great importance in reducing the operational energy demand of buildings. We developed a new kind of inorganic ...

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High thermoelectric performance in metal phosphides MP 2 (M= Co, Rh and Ir): a theoretical prediction from first-principles calculations

CJ Kang, UG Jong, YH Kye, CJ **Yu** - RSC advances, 2022 - pubs.rsc.org

Although metal phosphides have good electronic properties and high stabilities, they have been overlooked in general as thermoelectrics based on expectation of high thermal ...

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Influence of M/A substitution on material properties of intermetallic compounds MSn 2 (M= Fe, and Co; A= Li, and Na): a first-principles study

CJ **Yu**, US Hwang, YC Pak, K Rim, C Ryu... - New Journal of ..., 2020 - pubs.rsc.org

Iron and cobalt distannides MSn2 (M = Fe, and Co) are regarded as a promising conversion-type anode material for lithium- and sodium-ion batteries, but their properties are not well ...

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Contrary Effect of B and N Doping into Graphene and Graphene Oxide Heterostructures with MoS₂ on Interface Function and Hydrogen Evolution

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Molybdenum disulfide (MoS 2) attracts attention as a highly efficient and low-cost photocatalyst for hydrogen production but suffers from low conductance and high recombination rate ...

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Influence of low content metal intermixing on electrochemical stability and sodium ion transport in Na2 (Mg2– xZnx) TeO6 (x= 0.125, 1.875) for solid state sodium-ion ...

KC Ri, TI Ri, KM Kim, SH Choe, CJ **Yu** - Solid State Ionics, 2022 - Elsevier

Based on solid electrolyte instead of traditional liquid electrolyte, all-solid-state battery technology leads to the next generation of high performance sodium ion batteries. In this work, we ...

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Ab Initio Thermodynamic Study of PbI₂ and CH₃NH₃PbI₃ Surfaces in Reaction with CH₃NH₂ Gas for Perovskite Solar Cells

..., UG Jong, X Wang, Y Zhang, X Niu, CJ **Yu** - The Journal of ..., 2022 - ACS Publications
Hybrid organic–inorganic halide perovskites for photovoltaics have attracted research interest due to their unique material properties, but suffer from poor material stability. In this work, ...
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Mechanism of ultraviolet upconversion luminescence of Gd³⁺ ions sensitized by Yb³⁺-clusters in CaF₂: Yb³⁺, Gd³⁺

C Sin, T Aidilibike, W Qin, CJ **Yu** - Journal of Luminescence, 2018 - Elsevier
Ultraviolet (UV) luminescence under the near infrared (NIR) excitation is very important for the photodynamic therapy and NIR photocatalysis. In this paper, UV emission of Gd ³⁺ ions in ...
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Performance improvement of hole-transport material-free mesoporous perovskite solar cells with carbon electrodes using a solid–gas reaction

..., JH Ri, HS So, CI So, YS Kim, CJ **Yu** - ACS Applied Energy ..., 2021 - ACS Publications
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Improving the stability of hybrid perovskite FAPbI₃ by forming 3D/2D interfaces with organic spacers

YS Kim, CH Ri, YH Kye, UG Jong, CJ **Yu** - Chemical Communications, 2022 - pubs.rsc.org
... Yun-Sim Kim , Chol-Hyok Ri , Yun-Hyok Kye , Un-Gi Jong ORCID logo and **Chol-Jun Yu** ...
Chol-Jun Yu developed the original project and supervised the work. All the authors reviewed ...
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Revealing the mechanism of graphene oxide reduction by supercritical ethanol with first-principles calculations

JS Kim, CJ **Yu**, KC Ri, SH Choe... - The Journal of Physical ..., 2019 - ACS Publications
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First-principles study of the structural and electrochemical properties of Na_xTi₂O₄ (0≤ x≤ 1) with tunnel structure for anode applications in alkali-ion batteries

SH Choe, CJ **Yu**, YC Pak, YG Choe, KI Jon... - Physical Chemistry ..., 2021 - pubs.rsc.org
Due to their low cost and easy synthesis method, several kinds of sodium titanates have been explored as anode materials for sodium ion batteries (SIBs). However, some of them have ...
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First-principles study of mercaptoundecanoic acid molecule adsorption and gas molecule penetration onto silver surface: an insight for corrosion protection

CH Kim, C Ryu, YH Ro, CJ **Yu** - RSC advances, 2023 - pubs.rsc.org
... Chol-Ryu and **Chol-Jun Yu** performed the calculations and drafted the first manuscript.
Chung-Hyok Kim, Yong-... **Chol-Jun Yu** supervised the work. All authors reviewed the manuscript. ...
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Simulation of adsorption processes on the glass surface in aqueous solutions containing oxalic acid

J Kundin, CJ **Yu**, R Conradt, H Emmerich - Computational materials ..., 2010 - Elsevier
In this study we develop a kinetic model for the simulation of adsorption processes on a glass surface in the presence of oxalic acid, which is subsequently used for an investigation of ...
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Ab initio modeling of glass corrosion: hydroxylation and chemisorption of oxalic acid at diopside and åkermanite surfaces

CJ **Yu**, J Kundin, S Cottenier, H Emmerich - Acta materialia, 2009 - Elsevier
Using ab initio density functional theory, we have performed a systematic study of corrosion processes at pure and at hydroxylated surfaces of the silicate minerals diopside (CaMgSi₂ ...
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Ab initio insight into furan conversion to levulinate ester in reaction with methylal and methanol

YS Kim, RW Ham, YC Pak, CJ **Yu** - Materials Advances, 2024 - pubs.rsc.org
... Ryong-Wan Ham and **Chol-Jun Yu** developed the original project. Yun-Sim Kim and **Chol-Jun Yu** performed the ... **Chol-Jun Yu** supervised the work. All authors reviewed the manuscript. ...
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First-principles study of luminescence properties of the Eu-doped defect pyrochlore oxide $\text{KNbWO}_6 \cdot \text{H}_2\text{O}:0.125\text{Eu}^{3+}$

SH Choe, CJ **Yu**, M Choe, YH Kye, YN Han, G Pang - Physical Review B, 2020 - APS

Defect pyrochlore oxides have attracted great interest as a promising luminescent material due to their flexible composition and high electron/hole mobility. In this paper, we investigate ...

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Characterization of compositional particles in eye shadow powder by scanning electron microscope and X-ray mapping

UC Kim, KS Ju, CJ **Yu**, SH Kim, SP Ri - X-Ray Spectrometry, 2022 - Wiley Online Library

We developed an analysis method to evaluate the size and shape of particles of eye shadow cosmetics by applying the scanning electron microscopy (SEM) combined with the energy-...

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Atomistic simulations for material processes within multiscale method

CJ **Yu** - 2009 - researchgate.net

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Design of Improved Cathode Materials by Intermixing Transition Metals in Sodium-Iron Sulphate and Sodium Manganate for Sodium-Ion Batteries

CJ **Yu**, UG Jong, YH Kye, YS Kim - Computational Design of Battery ..., 2024 - Springer

Lithium-ion batteries (LIBs) are most widely used in portable electronic devices and often used to power hybrid and electric vehicles due to their high specific capacity and high energy ...

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Sodium Intercalation into Graphite and Graphene Complexes Towards Advanced Sodium-Ion Battery Anode Materials

SH Choe, KC Ri, SN Hong, C Ryu, CJ **Yu** - Computational Design of ..., 2024 - Springer

Lithium ion batteries (LIBs) are currently ubiquitous in almost all the products from portable electronics like mobile phones and computers to electric vehicles like electromotive cars or ...

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Structural and optoelectronic properties of the inorganic perovskites AGeX_3 (A= Cs, Rb; X= I, Br, Cl) for solar cell application

UG Jonga, CJ **Yu**, YH Kye, YG Choe, H Wei... - arXiv preprint arXiv ..., 2018 - arxiv.org

We predict the structural, electronic and optic properties of the inorganic Ge-based halide perovskites AGeX_3 (A = Cs, Rb; X = I, Br, Cl) by using first-principles method. In particular, ...

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Interface Engineering in Hybrid Iodide $\text{CH}_3\text{NH}_3\text{PbI}_3$ Perovskite Using Lewis Base and Graphene towards High Performance Solar Cells

CJ **Yu**, YH Kye, UG Jong, SG Ko, KC Ri... - arXiv preprint arXiv ..., 2019 - arxiv.org

Perovskite solar cells have achieved a substantial breakthrough via advanced interface engineerings. Reports have emphasized that combining the hybrid perovskites with Lewis base ...

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Supporting Information—First-principles study of sodium adsorption on defective graphene under propylene carbonate electrolyte conditions

C Ryu, SB Rim, Y Kang, CJ **Yu** - pdfs.semanticscholar.org

Figure S3. Isosurface view of the electronic density differences at the value of $0.0015|e|/\text{\AA}^3$ upon Na adsorption at the N1 hollow sites over the defective graphene with (a) MV,(b) DV, ...

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Table S1. Löwdin charge of atoms in $\text{Na}_2\text{TiO}_3\text{O}_7$ (NTO-2), $\text{Na}_4\text{Ti}_3\text{O}_7$ (NTO-4), $\text{Na}_2\text{Ti}_2\text{O}_7$ (NTNO-2), $\text{Na}_4\text{Ti}_2\text{O}_7$ (NTNO-4), $\text{Na}_2\text{Ti}_2\text{O}_7$ (NTVO-2) and ...

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Nature of Ionic Diffusion and Electronic Transfer in Eldfellite $\text{Na}_x\text{Fe}(\text{SO}_4)_2$

CJ **Yu**, SH Choe, GC Ri, SC Kim, HS Ryo... - arXiv preprint arXiv ..., 2016 - researchgate.net

Discovering new electrodes for sodium-ion battery requires clear understanding of the material process during battery operation. Using first-principles calculations, we identify ...

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